Research defense and next questions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Presentation and challenge session. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain the mechanism and terminology of research defense and next questions. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **research defense** and **next question**s.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Presentation and challenge session. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Independent reproduction check. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **research defense** and **next question**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Presentation and challenge session. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **research defense** and **next question**s using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Research defense and next questions case. Separate observations, given facts and assumptions before choosing an explanation.

A **research defense** distinguishes what was learned, what failed and which **next question** is justified. A complete investigation can end with a limited or **negative finding**.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Research defense and next questions. Connect each labelled input, deciding step and result.

Present the frozen question, credible baseline, unit design, primary analysis, uncertainty assumptions, failures and reproduction record. Propose next work that resolves a named gap with new **evidence**.

Research defense and next questions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Study shows candidate12/16 versus baseline11/16; replay matches; no fresh population sample or user study.	Present the frozen question, credible baseline, unit design, primary analysis, uncertainty assumptions, failures and reproduction record. Propose next work that resolves a named gap with new evidence.	75% versus68.75%, gain6.25 points in16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Research defense and next questions. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Study shows candidate12/16 versus baseline11/16; replay matches; no fresh population sample or user study. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: Change cameras as well as lighting in the follow-up.

Predicted result

Prediction: The design again entangles two conditions.

Why the result follows

Reason: A camera difference provides another explanation for any performance gap.

Words that unlock the idea

> Use the vocabulary of Research defense and next questions to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Research defense	A justified account of methods, findings and limits
Next question	A new bounded inquiry motivated by a gap
Negative finding	A result failing the declared expectation

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Research defense and next questions, predict a step, then compare with its explanation.

Study shows candidate12/16 versus baseline11/16; replay matches; no fresh population sample or user study.

75% versus68.75%, gain6.25 points in16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

Worked explanation · Research defense and next questions

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Research defense and next questions

Defend the next research question. A research defence identifies an observed weakness and a testable follow-up.

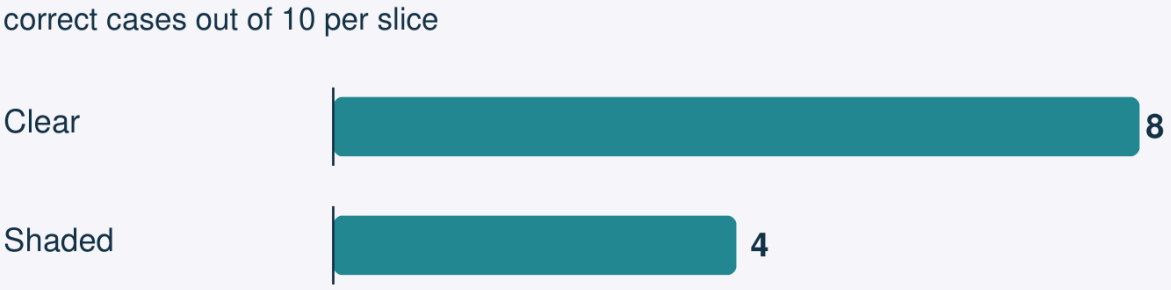
The facts we will use

Supplied facts

A frozen model is correct on 8/10 clear images and 4/10 shaded images. Source objects are grouped within each condition.

AI ACADEMY / AI-7/36

Defence: different condition slices



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/36

Research defense and next questions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State slice outcomes

Clear is 80%; shaded is 40%.





2. Limit the interpretation

Shade sensitivity is plausible, but camera or object differences may also matter.





3. Design follow-up

Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

CHANGE ONE THING

Change cameras as well as lighting in the follow-up.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. State slice outcomes	Clear is 80%; shaded is 40%.
2. Limit the interpretation	Shade sensitivity is plausible, but camera or object differences may also matter.
3. Design follow-up	Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A defence should explain results, uncertainty and what would discriminate between explanations next. A controlled paired follow-up is more informative than simply collecting extra easy images.

What this example does not establish

Keep the stated limit

Even a paired follow-up needs adequate independent objects and justified analysis; no causal conclusion is claimed yet.

Keep the stated limit

Read the labelled picture: Compute the clear and shaded rates and identify alternative explanations. How does the same-object, one-camera follow-up help, and why does changing cameras again reintroduce confounding?

Keep the stated limit

Change one thing: Change cameras as well as lighting in the follow-up. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Research defense and next questions claim and explain a repair using the supplied evidence.

The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Research defense and next questions with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Research defense** and **next question**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Research defense** and **next question**s. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Compute the clear and shaded rates and identify alternative explanations. How does the same-object, one-camera follow-up help, and why does changing cameras again reintroduce confounding?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Research defense and next questions. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Study shows candidate12/16 versus baseline11/16; replay matches; no fresh population sample or user study. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	

Record	Your evidence
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Research defense and next questions. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: The **next question** follows **evidence** gaps; it must not be presented as an answer already obtained.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Project candidate9/12, baseline10/12; reproduction matches. State result and a useful **next question**.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Research defense and next questions case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Research defense and next questions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Study shows candidate12/16 versus baseline11/16; replay matches; no fresh population sample or user study.	Present the frozen question, credible baseline, unit design, primary analysis, uncertainty assumptions, failures and reproduction record. Propose next work that resolves a named gap with new evidence.	75% versus68.75%, gain6.25 points in16 cases; replay supports computational consistency. Broader usefulness and generalisation remain untested.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Project candidate9/12, baseline10/12; reproduction matches. State result and a useful **next question**.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Project candidate9/12, baseline10/12; reproduction matches. State result and a useful **next question**.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Research defense and next questions, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Defend the next research question. A research defence identifies an observed weakness and a testable follow-up.

By the end, I can	How I will show it
Explain the mechanism and terminology of research defense and next questions	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Research defense and next questions without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **research defense** and **next question**s

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Defend the next research question. A research defence identifies an observed weakness and a testable follow-up.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

1. Design follow-up

2. Limit the interpretation

3. State slice outcomes

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Design follow-up Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

A defence should explain results, uncertainty and what would discriminate between explanations next. A controlled paired follow-up is more informative than simply collecting extra easy images.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change cameras as well as lighting in the follow-up. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. The design again entangles two conditions.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The design again entangles two conditions. A camera difference provides another explanation for any performance gap.

Repair a missing building block

Review Presentation and challenge session (ai-7/35). Goal: Explain the **mechanism** and terminology of presentation and challenge session. Short practice: Answer the camera-B challenge using only the twelve camera-A records.

Why is one displayed camera-B success a demonstration rather than broad **evidence** of camera-B reliability?. Return to this week and explain what you changed.

Review Independent reproduction check (ai-7/33). Goal: Explain the **mechanism** and terminology of independent reproduction check. Short practice: Locate the Q3 scoring disagreement between 5/6 and 4/6.

What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?. Return to this week and explain what you changed.

Review Writing the methods section (ai-7/30). Goal: Explain the **mechanism** and terminology of writing the methods section. Short practice: Apply the inclusive four-character rule to oak, pine and willow.

Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

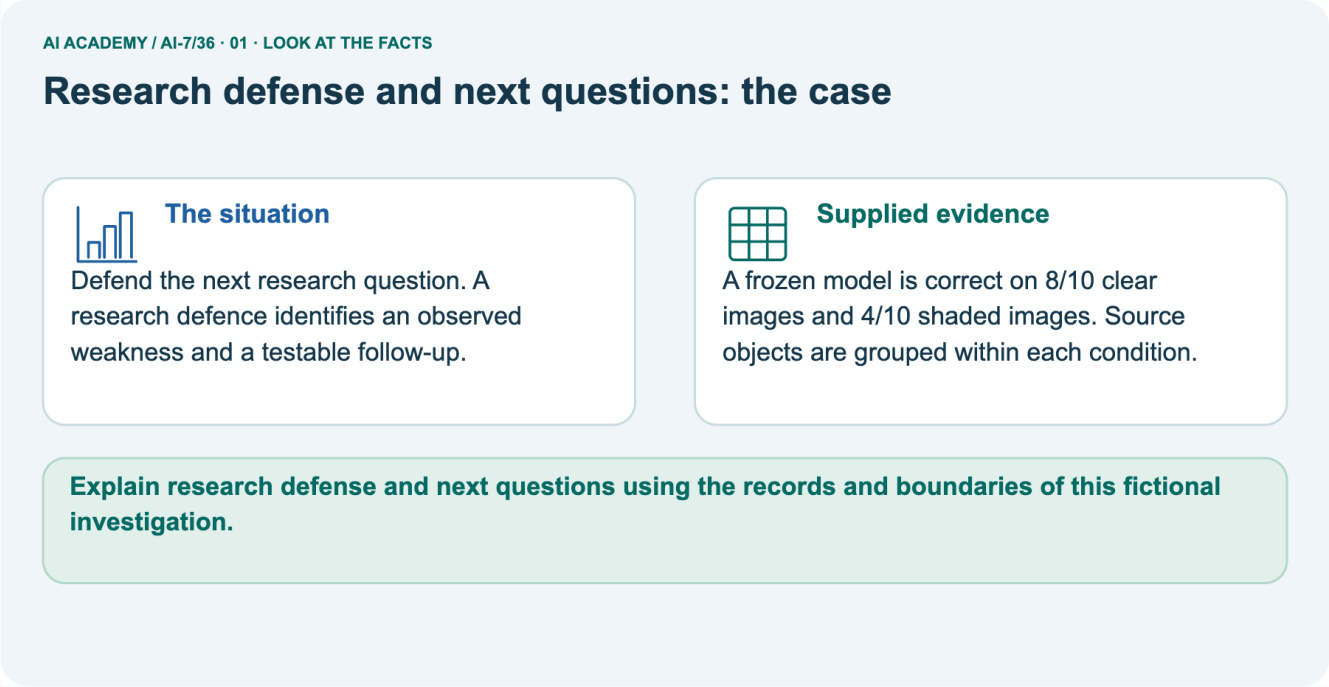
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Research defense and next questions graphic using its labels, deciding evidence and relevant limitation.

01. Research defense and next questions: the case



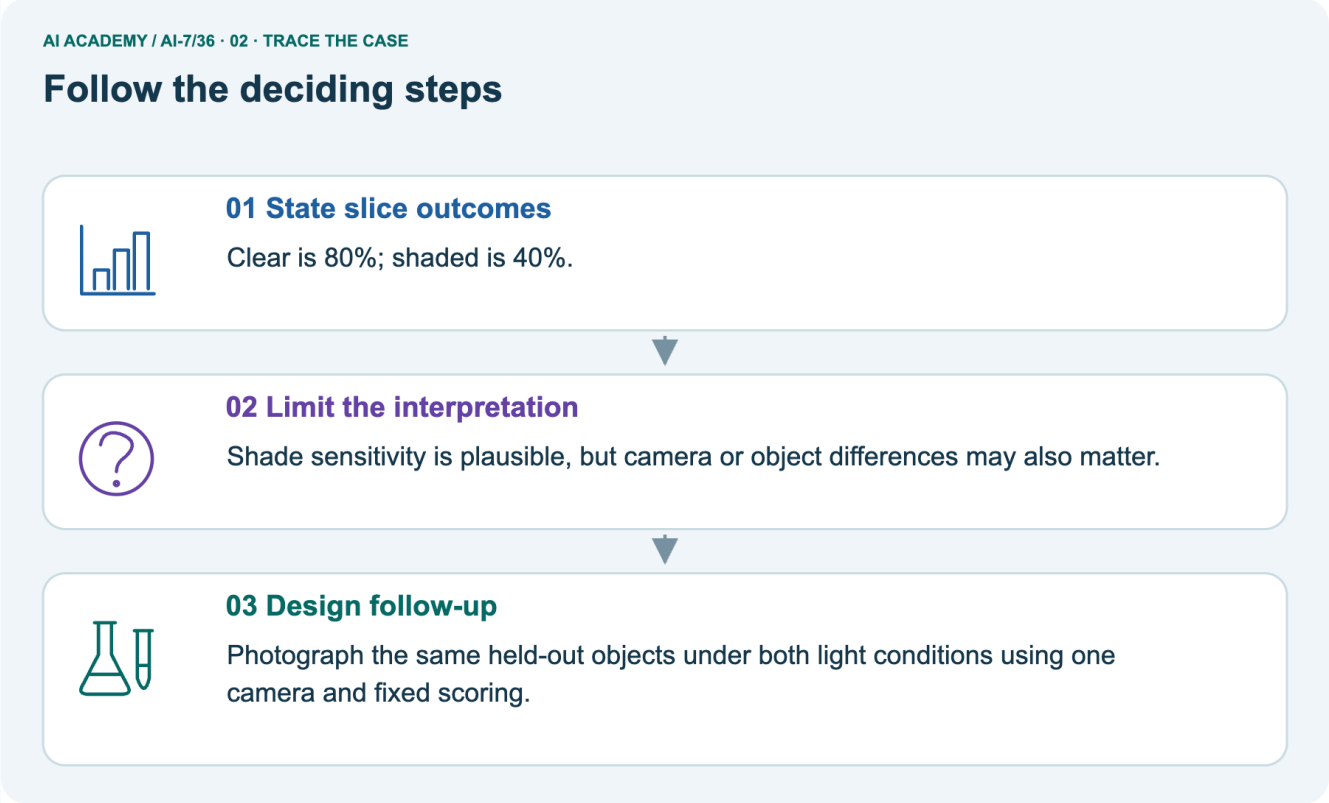
Follow the labelled picture with the matching explanation. Open the vector diagram

Defend the next research question. A research defence identifies an observed weakness and a testable follow-up. Supplied facts: A frozen model is correct on 8/10 clear images and 4/10 shaded images.

Source objects are grouped within each condition.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

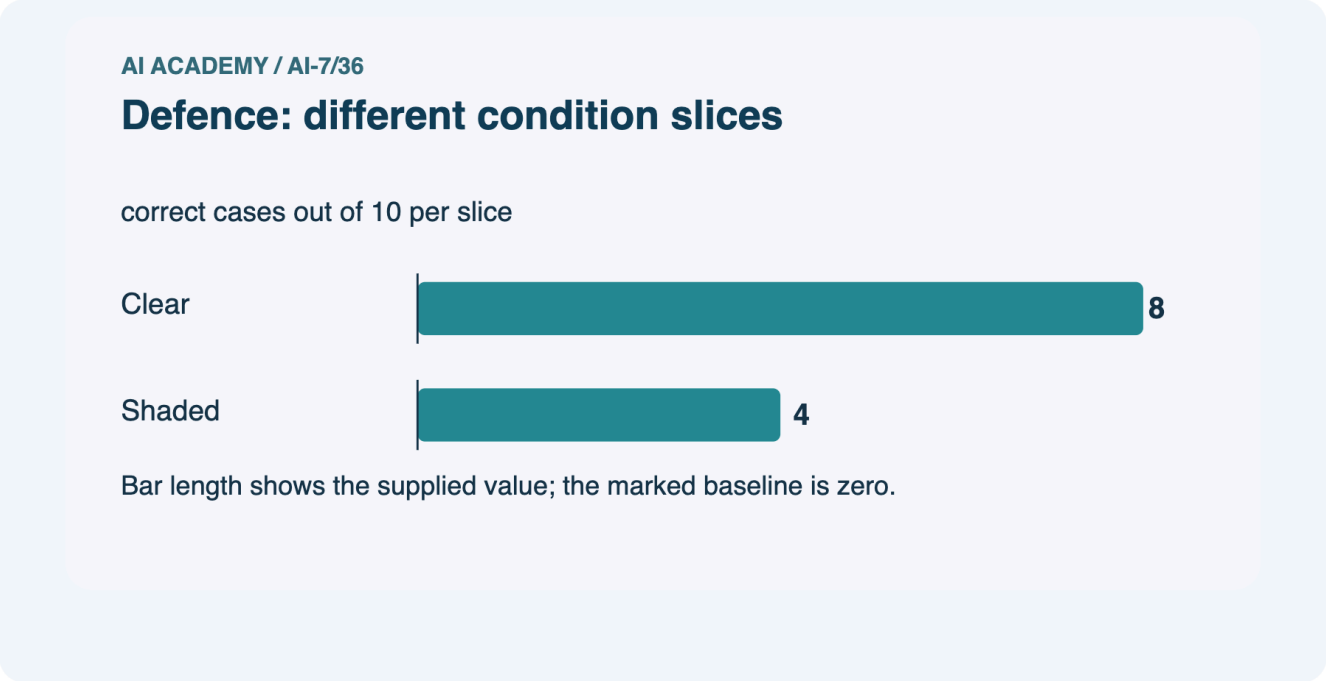


Follow the labelled picture with the matching explanation. Open the vector diagram

State slice outcomes: Clear is **80%**; shaded is 40%. → Limit the interpretation: Shade sensitivity is plausible, but camera or object differences may also matter. → Design follow-up: Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

Point to each numbered stage and explain which recorded **input** it uses.

03. Defence: different condition slices

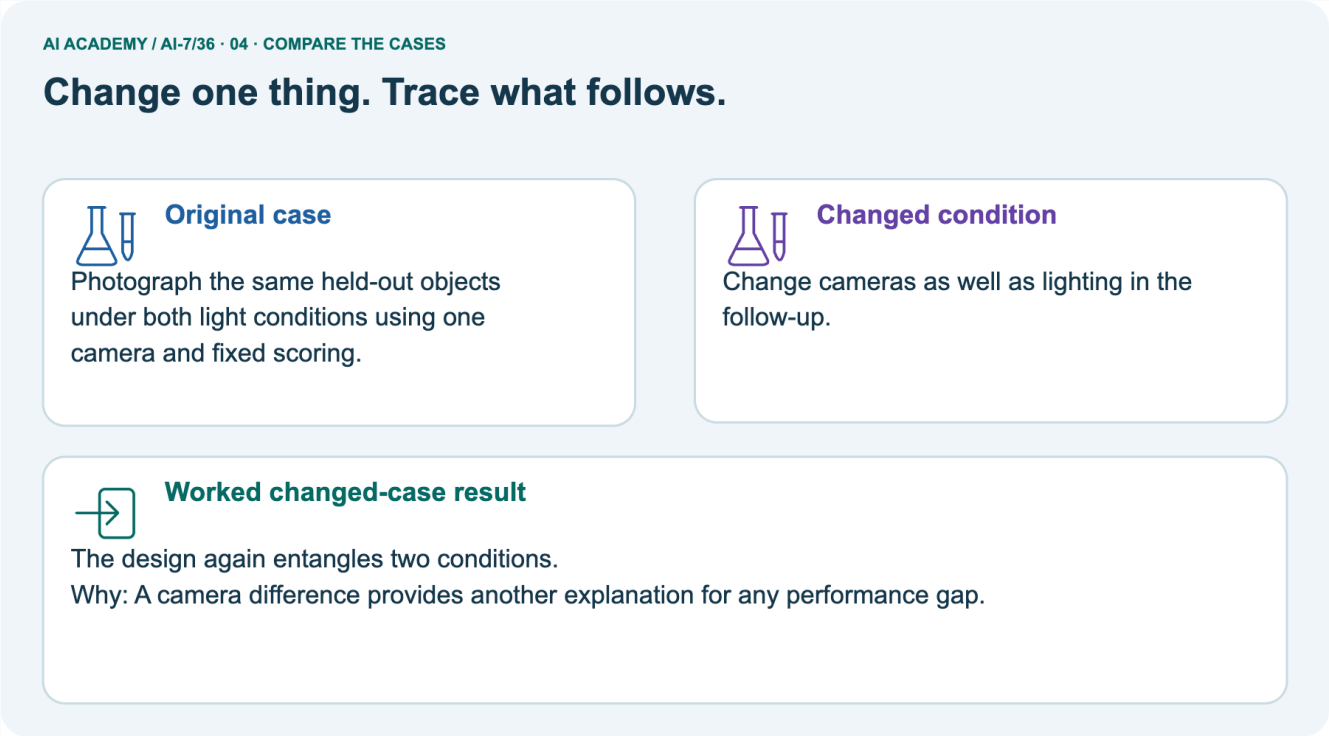


Follow the labelled picture with the matching explanation. Open the vector diagram

Defence: different condition slices. Units: correct cases out of 10 per slice. Clear: 8. Shaded: 4.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



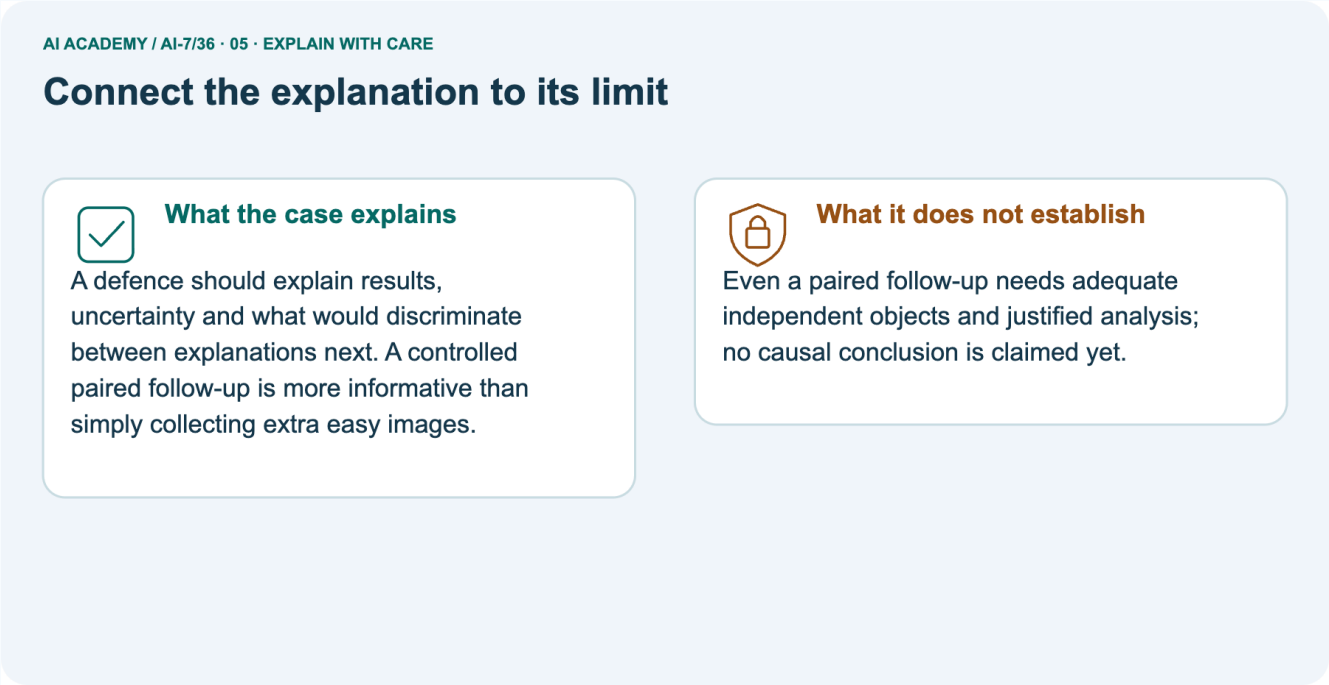
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Photograph the same held-out objects under both light conditions using one camera and fixed scoring. Changed condition: Change cameras as well as lighting in the follow-up.

Expected result: The design again entangles two conditions. Reason: A camera difference provides another explanation for any performance gap.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



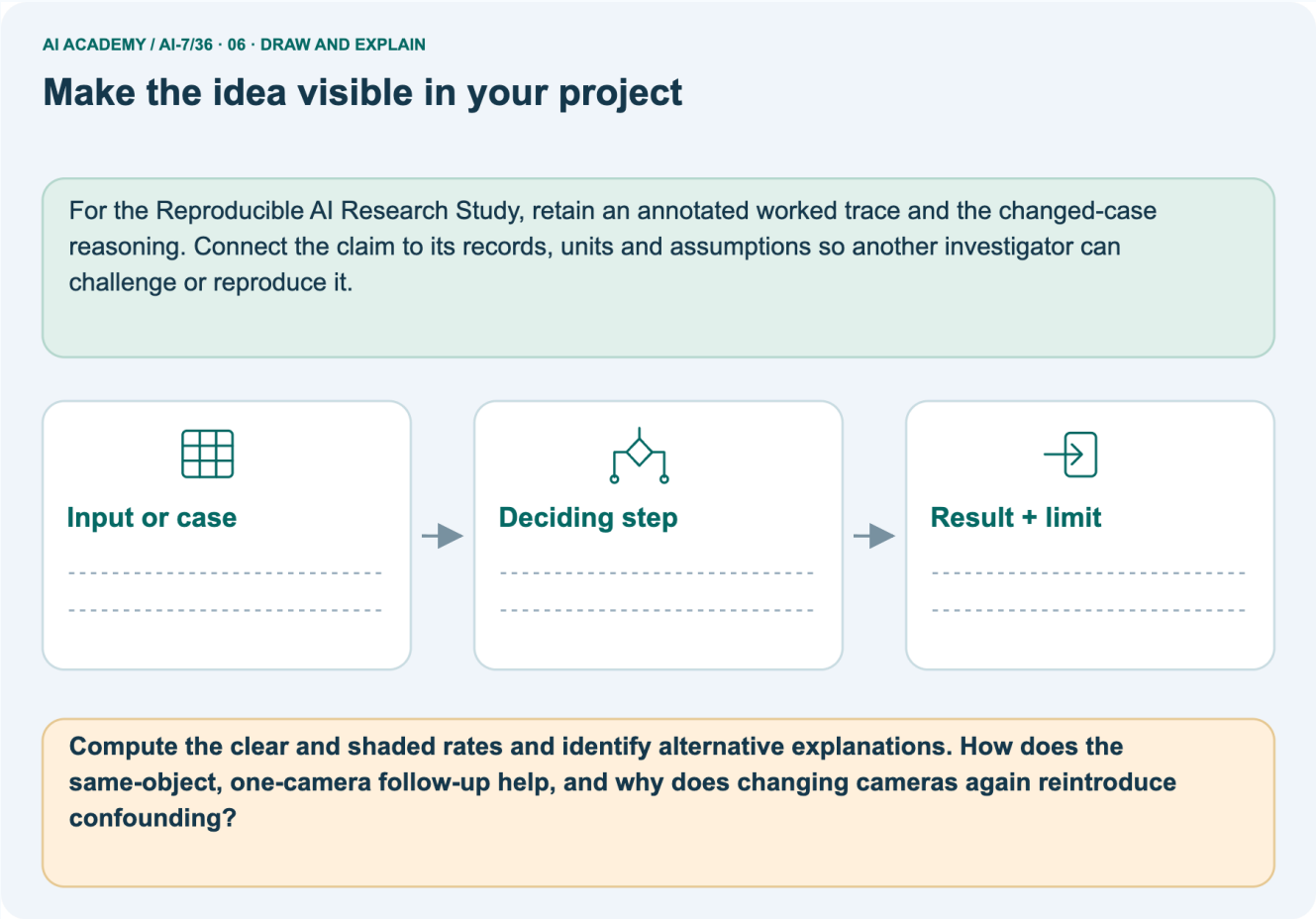
Follow the labelled picture with the matching explanation. Open the vector diagram

A defence should explain results, uncertainty and what would discriminate between explanations next. A controlled paired follow-up is more informative than simply collecting extra easy images.

Limit: Even a paired follow-up needs adequate independent objects and justified analysis; no causal conclusion is claimed yet.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Compute the clear and shaded rates and identify alternative explanations.

How does the same-object, one-camera follow-up help, and why does changing cameras again reintroduce confounding?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.